D3 Proposed Package Leaflet

Intermectin 10mg/ml Solution for Injection

Controlled Medicine
Keep out of reach of children
Jauhi dari kanak-kanak
MAL registration number

Composition:
Contains per ml:
Ivermectin ………………………………………………………….. 10mg
Glycerol formal ……………………………………………………….. 486mg
Propylene glycol ……………………………………………………… ad to volume

Product description:
Clear, colourless liquid.

Pharmacodynamics:
Ivermectin is a member of the macrocyclic lactone class of endectocides, which have a unique mode of action. Compounds of the class bind selectively and with high affinity to glutamate-gated chloride ion channels, which occur in invertebrate nerve and muscle cells. This leads to an increase in the permeability of the cell membrane to chloride ions with hyperpolarization of the nerve or muscle cell, resulting in paralysis and death of the parasite. Compounds of this class may also interact with other ligand-gated chloride channels, such as those gated by the neurotransmitter gamma-aminobutyric acid (GABA).
The margin of safety for compounds of this class is attributable to the fact that mammals do not have glutamate-gated chloride channels, the macrocyclic lactones have a low affinity for other mammalian ligand-gated chloride channels and they do not readily cross the blood-brain barrier.

Pharmacokinetics:
Maximum plasma concentration
Cattle
At a dose level of 0.2 mg ivermectin per kg a maximum plasma concentration of 35-50 ng/ml is reached in about 2 days and the half-life in plasma is 2.8 days. It is also established that ivermectin is carried mainly in the plasma (80%). This distribution between plasma and blood cells remains relatively constant.

Sheep
At a dose level of 0.3 mg ivermectin per kg an average peak of 16 ng/ml is reached one day after injection.

Pig
During trials carried out at a dose rate of 0.2 mg/kg ivermectin, a plasma concentration of 10-20 ng/ml was reached in about 2 days and half-life in plasma was 0.5 days.

Excretion: length of time and route
Cattle
In cattle receiving a single dose of tritium-labelled ivermectin, faeces collected during the first 7 days after dosing contained almost all the dosed radioactivity. Only about 1-2 % of the dosed radioactivity was excreted in the urine. Analyses of the faeces showed that about 40-50% of the excreted radioactivity was present as unaltered drug. The remaining 50-60% was present as metabolites or degradation products almost all which were more polar than the ivermectin.

Sheep
Radioactive ivermectin was administered to sheep at a dose rate of 0.3 mg per kg. Analyses of the faeces showed that about 99% of the drug and its metabolites are excreted in the faeces, about 1 % being excreted in the urine.

Pig
In pigs receiving a single dose of tritium-labelled ivermectin, faeces collected during the first 7 days after dosing contained only about 36% of the dosed radioactivity. Less than 1% of the dosed radioactivity was found in the urine. Analysis of the faeces showed that about 40% of the excreted radioactivity was unaltered drug.

Indication:
Indicated for the treatment and control of the following parasites in cattle, sheep and pigs:

Cattle:

Gastrointestinal nematodes (Adult & L4)
Ostertagia ostertagi(including inhibited larvae)
Ostertagia lyrata
Haemonchus placei
Trichostrongylus axei
Trichostrongylus colubriformis
Cooperia oncophora
C. punctata
C. pectinata
Cooperia spp.
Oesophagostomum radiatum
Nematodirus helvetianus (adult)
N. spathiger(adult)

Strongyloides papillosus (adult)
Bunostomum phlebotomum
Trichuris spp. (adult)
Lungworms (adult, L4, & inhibited larvae)
Dictyocaulus viviparus
Other nematodes
Thelazia spp. (adult)
Toxocara vitulorum(adult)
Roundworms
Parafilaria bovicola
Warbles (parasitic stages)
Hypoderma bovis
Hypoderma lineatum
Lice
Linognathus vituli
Haematopinus eurysternus
Solenopotes capillatus
Mites
*Psoroptes ovis (*syn. *P. communis* var. *bovis)*
Sarcoptes scabiei var.*bovis*

This product helps to control:

Lice
Damalinia bovis
Mites
Chorioptes bovis

This product, administered at the recommended dosage of 1 ml per 50 kg bodyweight, controls infections with *Haemonchus placei* and *Cooperia* spp. during the first 14 days after treatment, as well as acquired infections with *Ostertagia ostertagi* and *Oesophagostomum radiatum* during the first 21 days after treatment and acquired infections with *Dictyocaulus viviparus*during the first 28 days after treatment.

<u>**Sheep:**</u>

Gastrointestinal nematodes (Adult & L4)
Haemonchus contortus
Ostertagia circumcincta
O. trifurcata
Trichostrongylus axei (adult)
T. colubriformis
T. vitrinus (adult)
Nematodirus filicollis
N. spathiger (l4)
Cooperia curticei
Oesophagostomum columbianum
O. venulosum (adult)
Strongyloides papillosus (l4)
Gaigeria pachyscelis
Lungworms
Dictyocaulus filaria
Protostrongylus rufescens (adult)
Warbles (all larval stadia)
Oestrus ovis
Mites
Psoroptes ovis (P. communis var.*ovis)*
Sarcoptes scabiei
Psorergates ovis

<u>**Pig:**</u>

Gastrointestinal nematodes
Ascaris suum (adult & L4)
Hyostrongylus rubidus (adult & L4)
Oesophagostomum spp. (adult & L4)
Strongyloides ransomi (adult) *
Trichuris suis (adult)
Lungworms
Metastrongylus spp. (adult)
Lice

Haematopinus suis
Mites
Sarcoptes scabiei var.*suis*

***** Treat sows 7 - 14 days prior to farrowing to prevent infections with *Strongyloides ransomi* in piglets effectively through the milk.

Recommended dose:
Young and adult animals

Cattle:
This product should be given only by subcutaneous injection at the recommended dosage level of 0.2 mg ivermectin per kg bodyweight (corresponding to 1 ml per 50 kg bodyweight).
Inject under the loose skin of the shoulder. It is recommended to use a sterile needle of 15 to 20 mm. Sterile equipment should be used.

Treatment scheme in areas where hypodermosis occurs:
Ivermectin is highly effective against all stages of cattle grubs. However, a proper timing of the treatment is very important. To achieve optimal results, animals should be treated as soon as possible once heel fly season has ended. As is the case with other products for treatment of hypodermosis, the destruction of Hypoderma larvae may cause adverse reactions in the host when these larvae are situated in the vital parts of the animal. The destruction of *Hypoderma lineatum* may cause bloating when the larvae is located at the height of the submucosa of the oesophagus. The destruction of *Hypoderma bovis* may cause staggering or paralysis. Cattle should be treated before or after the presence of these larvae in vital parts of the body.Cattle, treated with this product at the end of the heel fly season, can be treated again during the winter with this veterinary medicine against internal parasites, scabies and lice without risk of reactions that are bound to the localization of the *Hypoderma* larvae.

Sheep:
A single administration should be done only by subcutaneous injection at the recommended dosage level of 0.2 mg ivermectin per kg bodyweight (corresponding to 0.5 ml per 25 kg bodyweight).
Inject under the loose skin behind the shoulder. In sheep with much wool you must ensure that the needle has penetrated the wool and skin before giving the dose. Use sterile equipment
For a complete treatment against scabies with *Psoroptes*, 2 injections with an interval of 7 days are necessary.
It is important to use the correct dose to reduce the risk of developing resistance.
Therefore, bodyweight should be determined as accurately as possible.

Pig:
Young and adult animals
This product should be given only by subcutaneous injection in the neck at the recommended dosage level of 0.3 mg ivermectin per kg bodyweight (corresponding to 1 ml per 33 kg bodyweight).
The injection may be administered with a single-dose syringe or automatic syringe equipment.
Work aseptically.

Breeding animals
At the time of initiating any parasite control program, it is important to treat all breeding animals in the herd. After the initial treatment, use the product regularly as follows:
Sows: Treat sows preferably 7 - 14 days prior to farrowing to prevent infection of the piglets.
Gilts: Treat 7-14 days prior to breeding or farrowing.
Boars: Frequency and need for treatments are dependent upon exposure to parasites and advice of the veterinarian.

Feeder pigs
All feeder pigs should be treated before placement in clean quarters. Pigs exposed to contaminated soil or pasture may need retreatment if reinfection occurs.
It is important to use the correct dose to reduce the risk of developing resistance.
To ensure administration of a correct dose, bodyweight should be determined as accurately as possible; accuracy of the dosing device should be checked.
If animals are to be treated collectively rather than individually, they should be grouped according to their bodyweight and dosed accordingly, in order to avoid under- or over- dosing.

Contraindications:
This product is not to be used intramuscularly or intravenously.
Do not use in case of hypersensitivity to the active substance.
Do not use in animal species other than target animals. Adverse reactions with fatal outcome were observed, usually in dogs - particularly in collies and Old English Sheepdogs.
Not approved for use in animals producing milk for human consumption. Do not use in dairy cows in dry conditions, including pregnant heifers, or in non-lactating ewes within 60 days of partus.

Warning and Precautions:
Special warnings for each target species
For an effective control of mite infestations, care must be taken to prevent reinfestation from exposure to untreated animals or contaminated facilities. Louse eggs are unaffected by ivermectin and may require up to 3 weeks to hatch. Louse infestations developing from hatching eggs may require retreatment.

Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

- Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.
- Underdosing, which may be due to underestimation of bodyweight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. *Faecal Egg Count Reduction Test*). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic, an anthelmintic belonging to another pharmacological class and having a different mode of action should be used.

Special precautions for use

Special precautions for use in animals

Do not use in animal species other than cattle, sheep and pigs.

In cats, dogs, especially Collies, Old English Sheepdogs and related breeds and crosses, and also in turtles/tortoises, side effects can be observed due to the concentration of ivermectin in this veterinary medicinal product. Side effects with fatal outcome were observed after administration of ivermectin to dogs - particularly in collies and Old English Sheepdogs and to turtles.

Do not combine the treatment with vaccination against lungworms. In case vaccinated animals should be treated, do not administer the product in a period of 28 days before or after vaccination.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

Do not smoke, drink or eat while handling the product.

Wash hands after use.

Avoid contact with eyes and skin. If this occurs, immediately rinse the affected areas with water.

In case of accidental self-injection, seek medical advice immediately and show the package leaflet or the label to the physician: the product may cause local irritation and/or pain at the site of injection.

Special precautions for the disposal of unused veterinary medicinal product or waste materials derived from the use of such products

Extremely dangerous to fish and aquatic life. Do not contaminate surface waters or ditches with product or used container. Any unused veterinary medicinal product or waste material derived from such veterinary medicinal product should be disposed of in accordance with local requirements.

Interaction with other medicaments:

This veterinary medicinal product can be used concurrently with foot and mouth disease vaccine or clostridial vaccine, given at separate injection sites.

In vitro, the activity of ivermectin has been shown to be increased by benzodiazepine derivatives.

Pregnancy and Lactation:

Pregnancy:

Studies have demonstrated a wide safety margin. At the recommended use level, no adverse effects on fertility or gestation in breeding animals were observed

Lactation:

Do not use in animals producing milk for human consumption. Do not use in non-lactating dairy cows, including pregnant heifers, or in non-lactating ewes within 60 days of partus.

Side Effects:

Mild and transient discomfort has occasionally been observed following subcutaneous administration. Very rarely allergic reactions and anaphylaxis can be observed. These can be accompanies by neurological symptoms such as ataxia, convulsion and / or tremor.

A low incidence of soft tissue swelling at the injection site has been observed.

In some sheep, discomfort, sometimes intense but usually transient, has been observed.

Symptoms and treatment of overdose:

Cattle

Single doses of 4.0 mg ivermectin per kg (20 x the use level) given subcutaneously resulted in ataxia and depression.

Sheep

Single doses of 4.0 mg ivermectin per kg (20 x the use level) given subcutaneously, resulted in ataxia and depression

Pig

A dose of 30 mg ivermectin per kg (100 x the recommended dose of 0.3 mg per kg) injected subcutaneously caused lethargy, ataxia, bilateral mydriasis, intermittent tremors, labored breathing and lateral recumbency.

No antidote has been identified; however, symptomatic therapy may be beneficial.

Withdrawal period:

Do not use in animals producing milk for human consumption.

Meat and offal:

Cattle: 49 days

Sheep: 22 days

Pig: 14 days

Do not use in non-lactating dairy cows, including pregnant heifers, within 60 days of calving.

Do not use in non-lactating ewes within 60 days of partus.

Storage condition:
Store below 30ºC. Do not refrigerate or freeze. Store in the original package. Protect from light.

Packaging:
Clear glass injection vial of 50ml and 100ml closed with a rubber stopper and an aluminium cap.

Manufacturer:
Interchemie werken “De Adelaar” Eesti AS
Vanapere tee 14, Püünsi village, Viimsi municipality, Harju country, 74013 Estonia.

Marketing Authorization Holder:
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